Discovery of Novel Tissue Culture Conditions to Expand the Growth of Primary Renal Epithelial Cells Cultured from the Urine of Children with Nephrotic Syndrome

CHRC Trainee Research Competition

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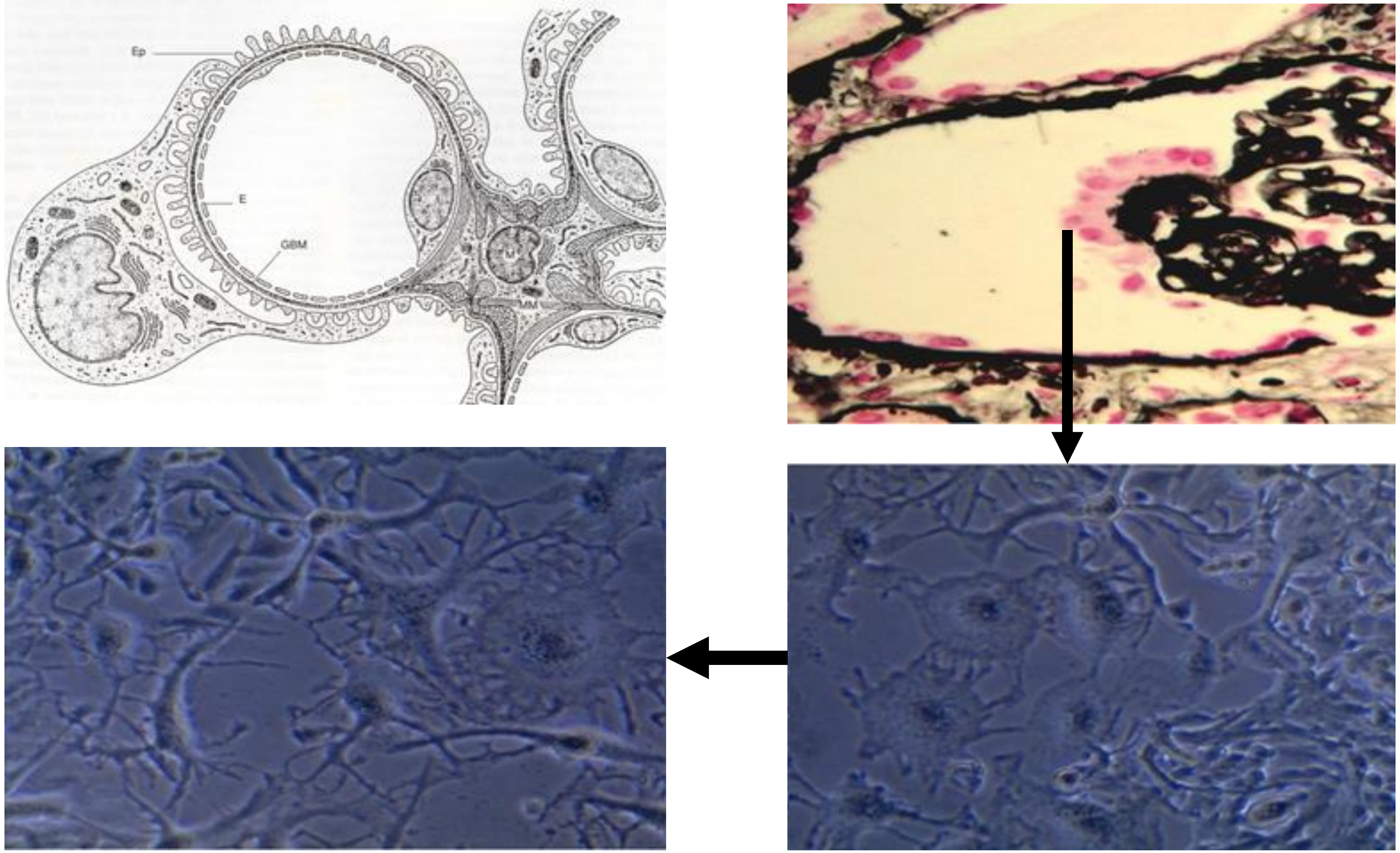
Background

- Podocyte injury plays a key role in the pathogenesis of nephrotic syndrome (NS). One limitation to study the pathogenesis (NS) in children is the lack of culture systems to expand podocytes and glomerular parietal epithelial cells (GPEc) from these patients.

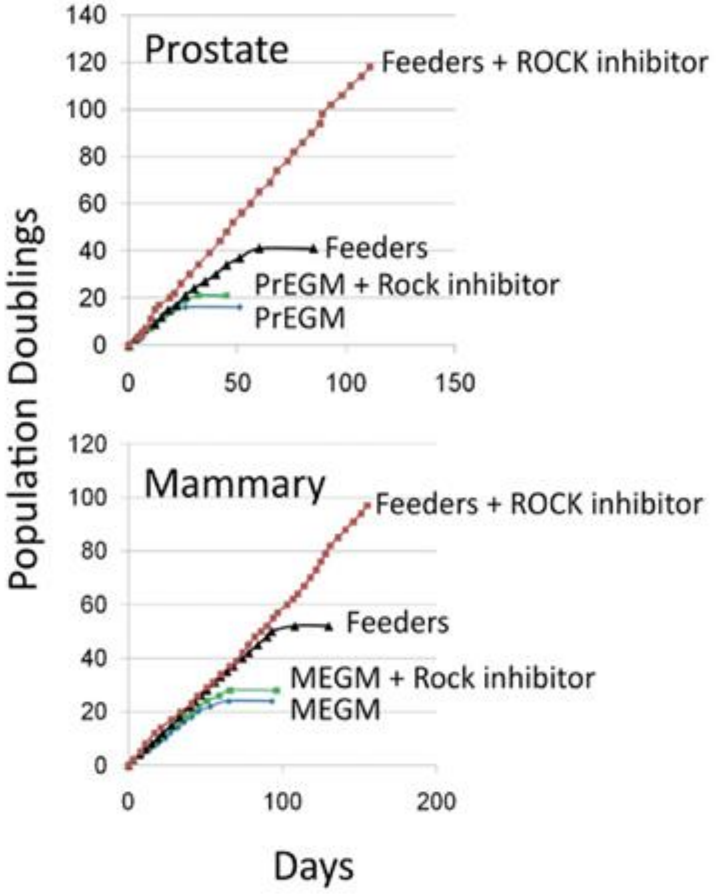


Background

- Conditional reprogramming medium (CRM) allows the unlimited proliferation of primary epithelial cells from many tissues without introducing genetic or viral manipulations.

- Fibroblast Growth Factor-2 (FGF-2) induces the proliferation of renal progenitor epithelial cells, podocytes, and tubular epithelial cells.

- It is unknown whether CRM or FGF-2 could facilitate the expansion and/or differentiation of primary podocytes and GPEc cultured from the urine of children with NS.



-Liu et al. (2012)

Objectives

- To develop tissue culture conditions that promote the conditional reprograming of podocytes and renal tubular epithelial cells cultured from the urine of children with nephrotic syndrome.
- To define the transcriptome profile changes associated with the conditional reprograming of podocytes, GPEc, and other urinary renal epithelial cells and define the role of FGF-2 in this process.

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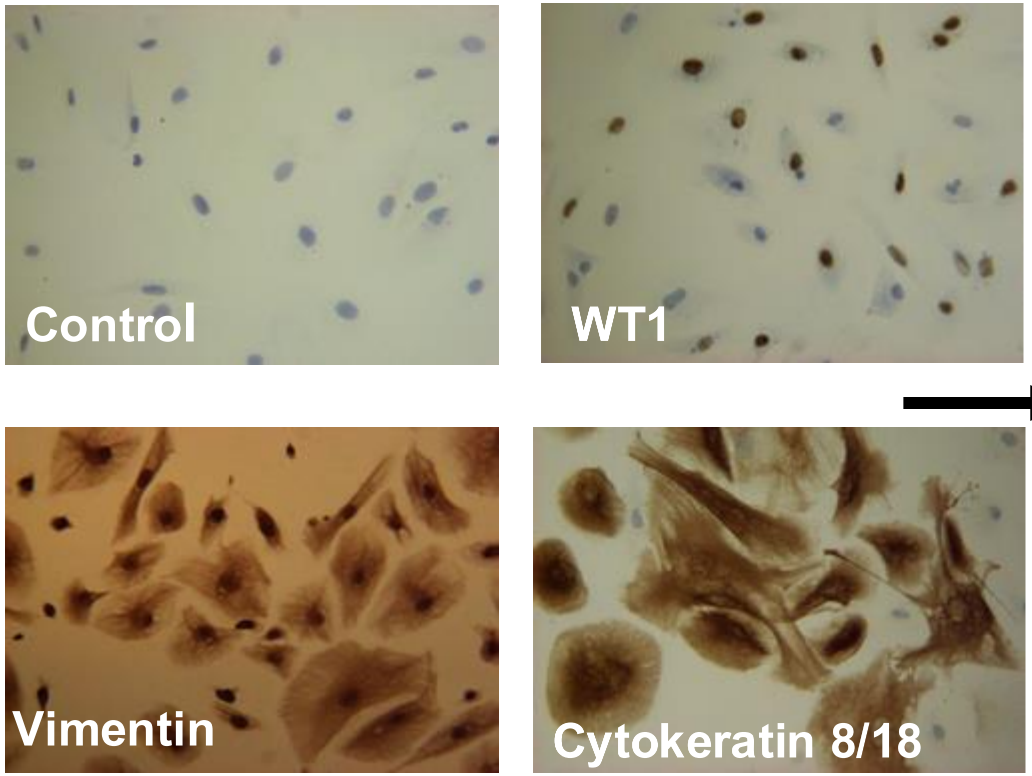
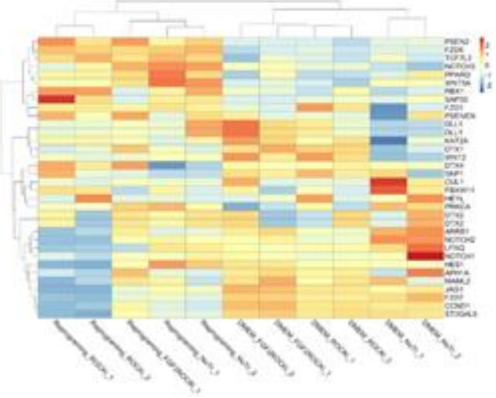
Methods and Experimental Design



Culture Conditions	Abbreviation
Control Media (DMEM + 10% FBS)	CM
CM + Rock inhibitor Y-27632 (10 µM)	CM+RI
CM + RI + FGF-2 (10 ng/ml)	CM+RI+FGF2
Reprograming Condition Medium (DMEM + condition medium from irradiated murine fibroblasts)[3]	CMIF
Conditional Reprograming Medium (CMIF+ RI)[1-3]	CRM
CRM + FGF-2 (10 ng/ml)	CRM+FGF2

Bulk RNA transcriptome profiling



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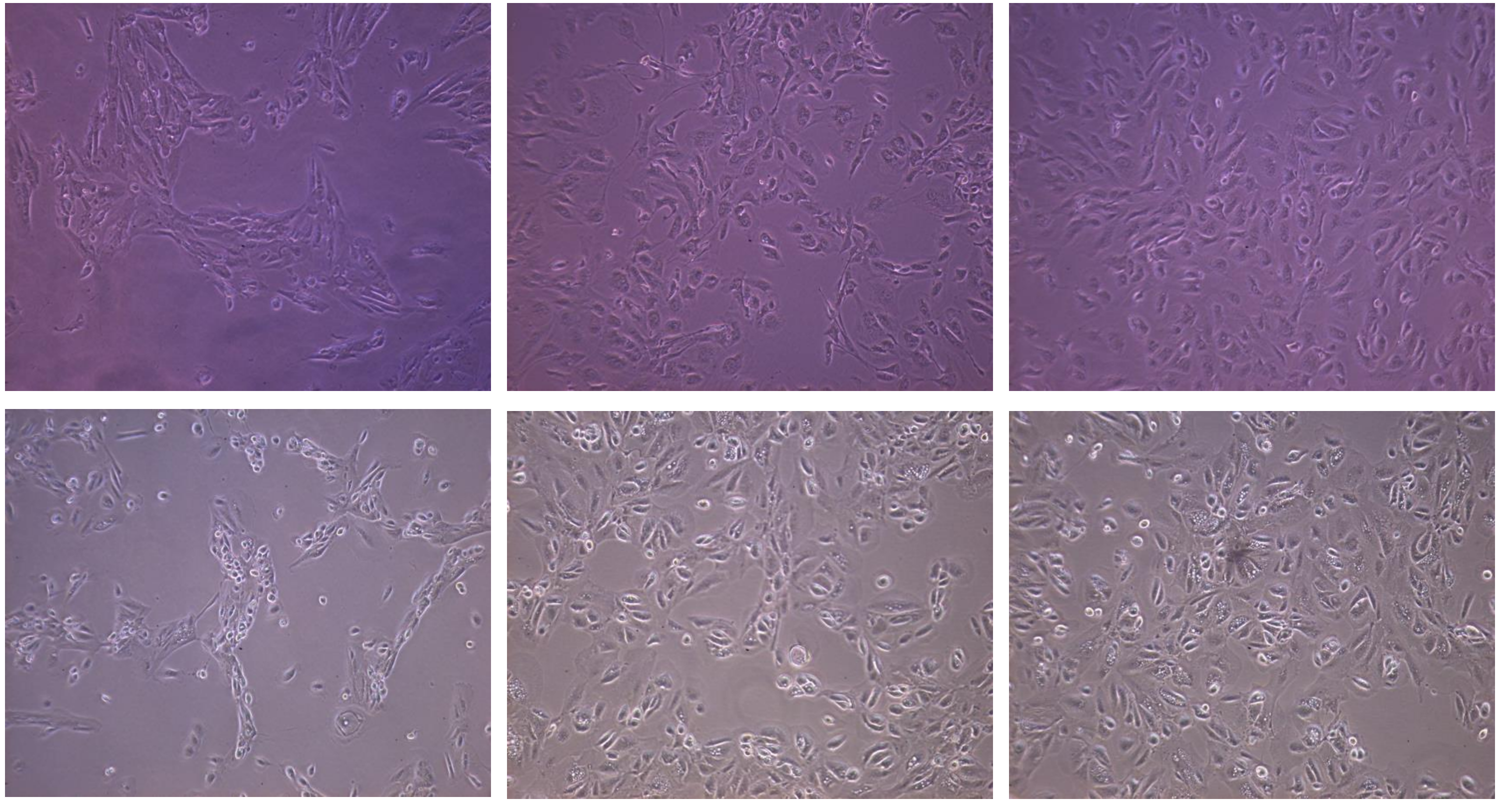
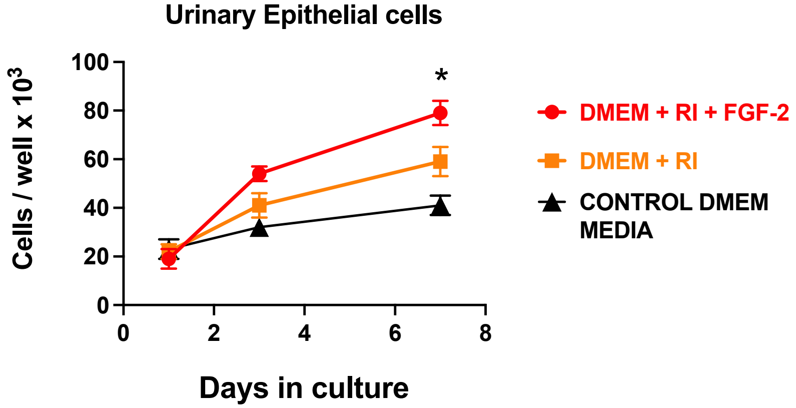
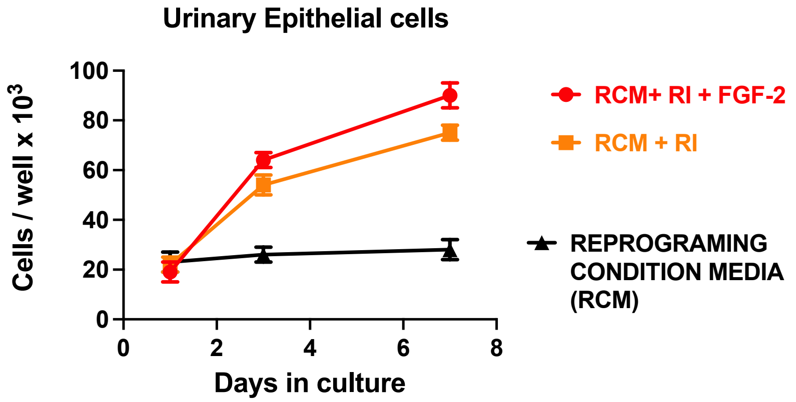
Proliferation







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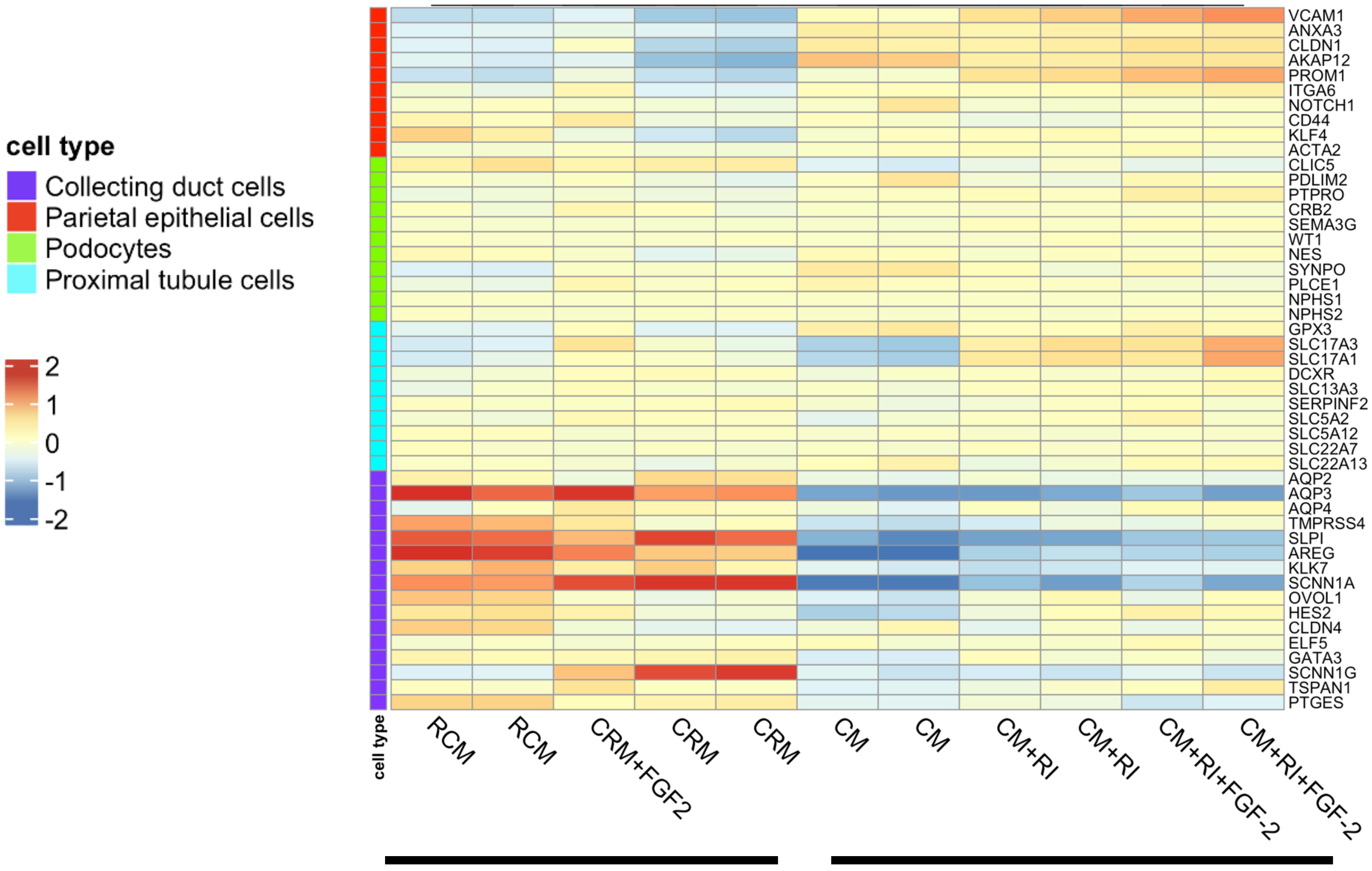
Bulk RNA-seq Expression Profile of UREC



Glomerular Parietal Epithelial cells/ Podocytes

Proximal Tubular Epithelial cells

Collecting Duct Epithelial cells

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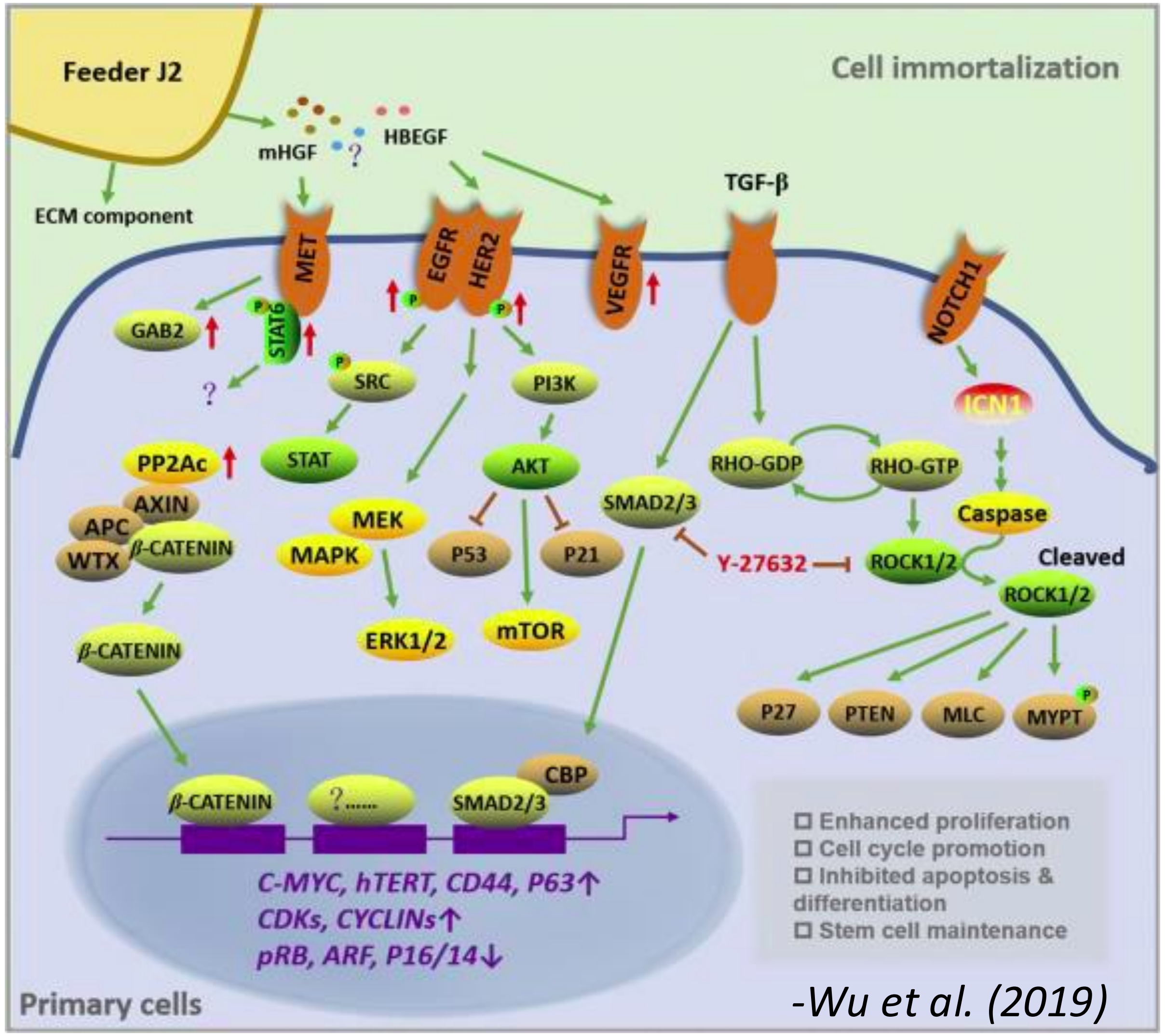
Mechanisms of Conditionally Reprogrammed UREC

MODEL: CRM (non-renal epithelial cells)



-Wu et al. (2019)

CRM EFFECTS ON CULTURED UREC

Proliferation: activates mainly the EGFR-MYBL2 pathway.

Inhibits the p53 pathway: which is critical to evade apoptosis.

Downregulates WT1 and EMT podocyte genes

Downregulates NOTCH1/2 signaling: NOTCH1 induces growth arrest, differentiation, and activates ROCK1.

ROCK inhibitors: RI suppress TGF-β/SMAD pathway and P27, PTEN, and the phosphatase myosin light chain

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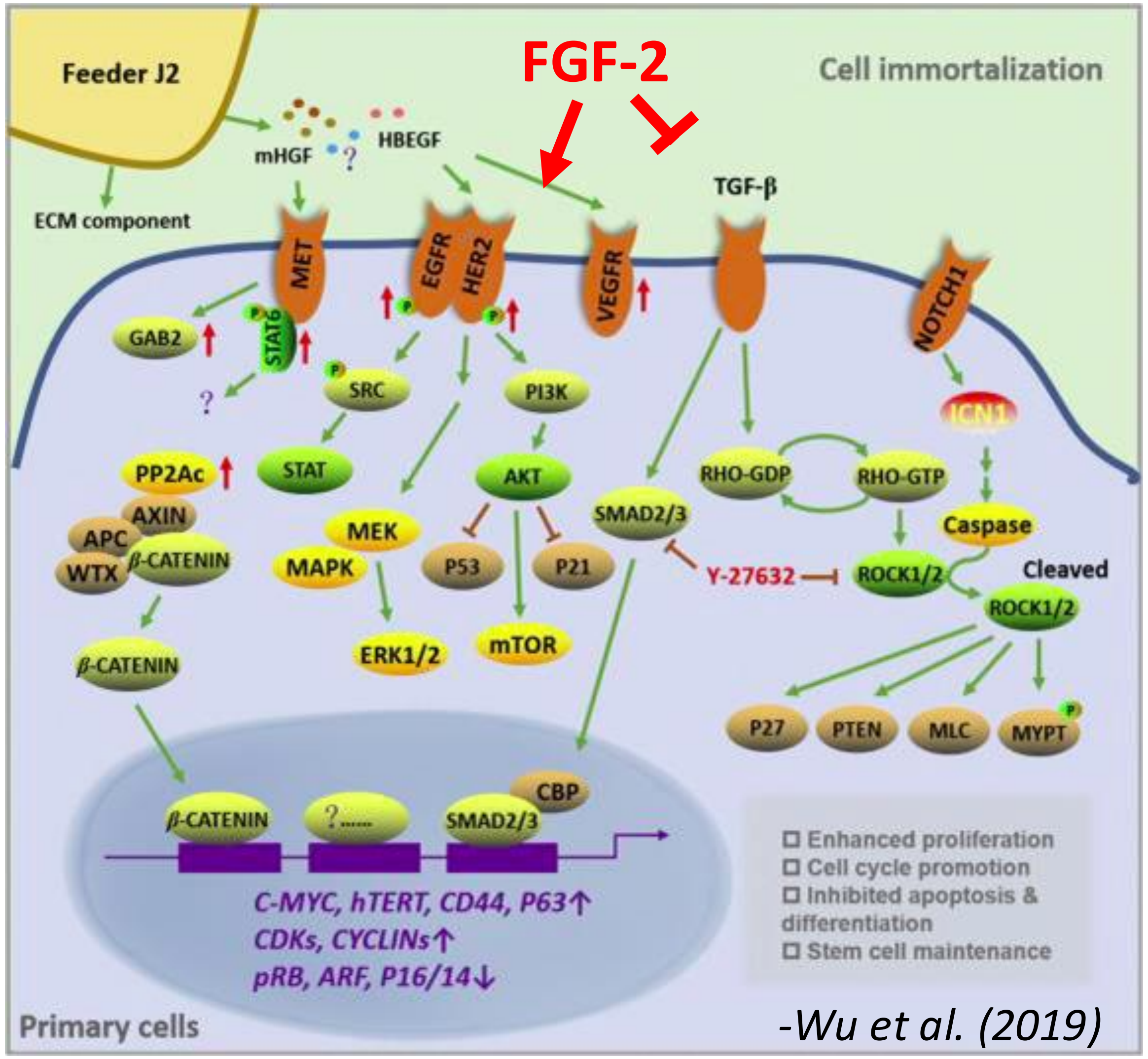
FGF-2 Effects on Primary UREC

MODEL: CRM (non-renal epithelial cells)



-Wu et al. (2019)

FGF-2 EFFECTS ON CULTURED UREC

ECM proteins: FGF-2 interacts with heparan sulfate proteoglycans facilitating cell to cell contacts and attachment.

Activates the Rho/ROCK/pMLC pathway, increasing contractility and differentiation of podocytes.

Apoptosis and stem cell markers: Inhibits apoptosis and induces the expression of the stem cell marker CD44

TGF-β signaling: Counteracts the growth inhibitor effects of the TGF–β/SMAD pathway.

Survival and proliferation: Induces SRC, PI3K, AKT, MAPK, ERK1/2, and interacts with the EGFR and MET pathways

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Conclusions

- We have identified tissue culture conditions that can sustain the growth of differentiated primary podocytes cultured from the urine of children with NS and have identified several signaling gene pathways that are involved in this process.

- Conditional reprograming medium (CRM) induces the expression of proliferating genes associated with the EGFR-MYBL2 pathway in cultured primary UREC, facilitating the reprograming and expansion of mainly collecting duct epithelial cells.

Thank you for your attention.

Future Directions

- Future studies need to be done to identify the tissue culture conditions that will promote the long-term expansion and differentiation of podocytes.
- We need to confirm that gene expression changes indeed reflect the cell type present in the tissue culture media
- These tissue culture conditions should be further validated by expanding UREC cultured from children with different clinical types of NS.